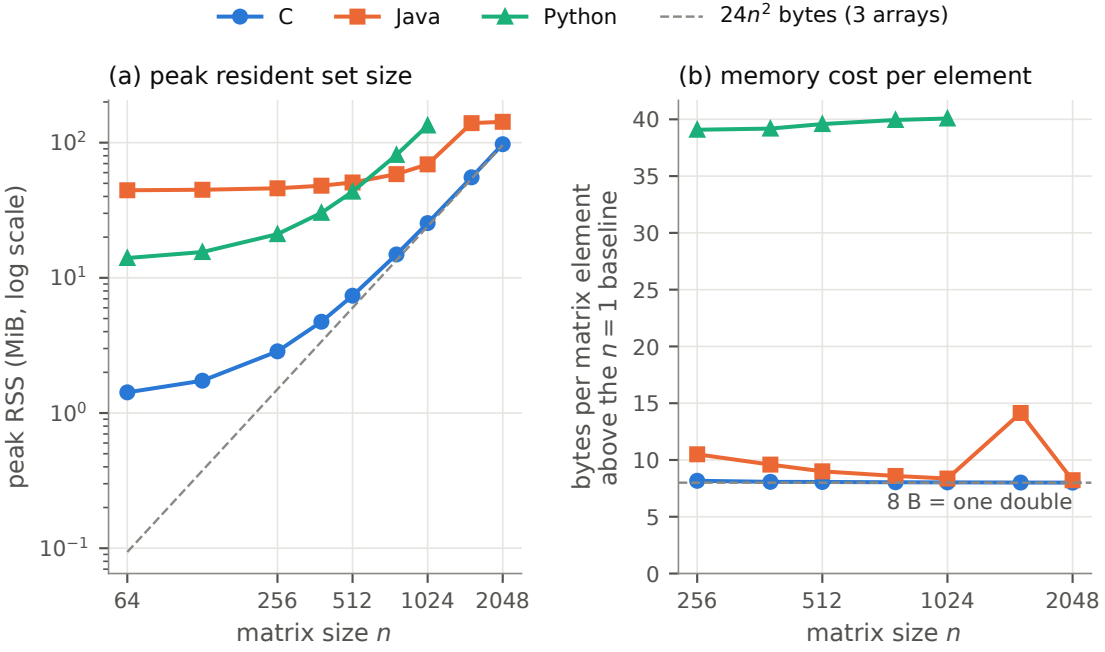
C
Java
Python
$24n^2$ bytes (3 arrays)
(a) peak resident set size
peak RSS (MiB, log scale)
10^2
10^1
10^0
10^{-1}
64
256
512
1024
2048
matrix size n
(b) memory cost per element
bytes per matrix element
above the $n = 1$ baseline
40
35
30
25
20
15
10
5
0
8 B = one double
256
512
1024
2048
matrix size n